

ggplot2

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This article discusses ggplot2, an open source R package, based on a grammatical theory of graphics. The underlying theory has been discussed in depth elsewhere so this article illustrates some of the consequences of the theory for creating new graphics, the importance of programmable graphics, and the rich ecosystem that has grown up around ggplot2. © 2011 John Wiley & Sons, Inc. *WIREs Comp Stat* 2011 3 180–185
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INTRODUCTION

ggplot2 is an open source R package that implements the layered grammar of graphics,¹ an extension of Wilkinson's grammar of graphics.² This article provides an overview of ggplot2 and the ecosystem that has built up around it. I will focus on the features that make ggplot2 different from other plot systems (the underlying theory and the programmable nature), as well as some of the important features of the community.

This article begins with a reminder about the motivation for visualization software, then continues to discuss three particularly special features of ggplot2: the underlying grammar, its programmable nature, and the ggplot2 community.

DATA ANALYSIS

When creating visualization software, it is useful to think about why we create visualizations: not to create pretty pictures, but to better understand our data. Visualization is just part of the data analysis process, as shown in Figure 1, and it needs to be coupled with transformation and modeling to build understanding. ggplot2 has been designed with this in mind. Because ggplot2 is embedded within R,³ we can use ggplot2 for visualization and other R packages can provide tools for transformation and modeling. All that is required is a common data format, and ggplot2 works with data in 'long' format, where variables are stored in columns and observations in rows. This means that you do not

need to change the format of your data as you iterate between modeling, transforming and visualizing.

A GRAMMAR OF GRAPHICS

Focusing on just the visualization component of the cycle, we ask two questions over and over again: what should we plot next and how can we make that plot? ggplot2 focuses on the second question: once you have come up with a plot in your head, how can you render it on screen as quickly as possible? Most graphics packages, like base graphics⁴ and lattice graphics⁵ in R, start with a posse of named graphics, like scatterplots, pie charts, and histograms, and a handful of primitives, like lines and text. To create a plot, you figure out the closest named graphic and then tweak plot parameters and add primitives to bring your idea to life. For complicated graphics, code is usually imperative: draw a line here, draw text there, do this, do that, and you have to worry about many low-level details.

If you are using a plotting system with an underlying grammar, such as ggplot2 or Wilkinson's GPL, you take a different approach. You think about how your data will be represented visually, then describe that representation using a declarative language. The declarative language, or grammar, provides a set of independent building blocks, similar to nouns and verbs that allow you to build up a plot piece by piece. You focus on describing what you want, leaving it up to the software to draw the plot.

Transitioning from the first approach to the second is often frustrating, because you have to give up much of the control that you are used to. It is much like learning LaTeX after learning MS Word: at first you are frustrated by how little control you have, but eventually the restrictions become freeing, leaving you to concentrate on the content, not the appearance.

Additional Supporting Information may be found in the online version of this article.

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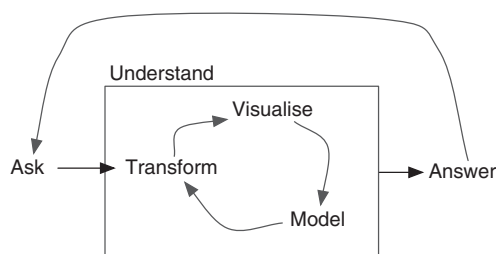


FIGURE 1 | The data analysis cycle.

Similarly, learning ggplot2 can be frustrating if you are familiar with other plotting systems, because controlling low-level aspects of plot appearance is considerably more difficult in ggplot2. However, the trade-off is worth it: once you give up this desire for low-level control, you can create richer graphics much more easily.

The following example gives a small flavor of ggplot2 and the grammar, by showing how to create a waterfall chart.

CASE STUDY: A WATERFALL CHART

The following example is inspired by an example from the Learn R blog^a. It shows how to create a waterfall chart, often used in business to display flows of income and expenses over time. Figure 2 shows a typical example, which we will recreate in two phases, first focusing on the essential structure of the plot, and then tweaking the appearance. This is a similar breakdown to exploratory versus communication graphics: first you figure out the best plot for the job with rapid iterations and once you have you spend more time polishing it for presentation to others.

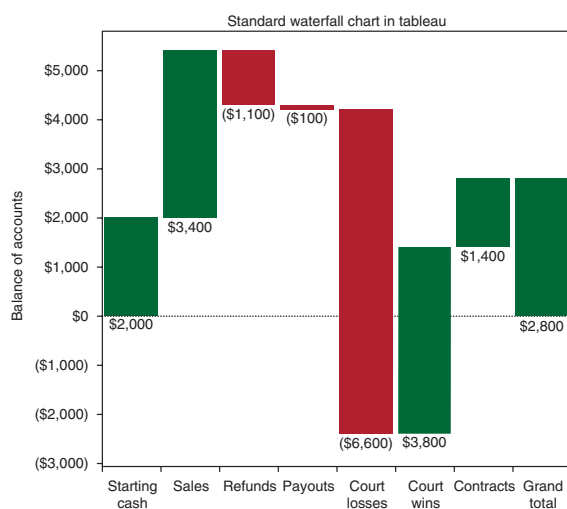


FIGURE 2 | A waterfall chart showing balance changes in a fiction company. Used with permission from Stephen McDaniel at Freakalytics.com.

To recreate this plot with ggplot2 we start by thinking about underlying data and how it is represented. What data does the plot display and how does it display it? The most striking feature is the rectangles that display the change in balance for each event. We could represent that data in R with the following data frame:

```
balance <- data.frame(event =
  c("Starting\nCash", "Sales",
    "Refunds",
    "Payouts", "Court\nLosses",
    "Court\nWins", "Contracts",
    "End\nCash"),
  change = c(2000, 3400, -1100, -100,
    -6600, 3800, 1400, -2800))
```

A little thought reveals an additional variable that we need: time, that is, what the x-axis displays, even though it is labeled with the event. If it is not clear why we need this extra variable, think about a company that received two payouts: we would not want to place them in the same position on the x-axis. It is also useful to add two other variables: the running balance, used to draw the starting point of the bar, and the direction of the flow, used to color the bars. ggplot2 does provide ways to do simple statistical transformations like this within the plot, but when creating a new plot it is often better to start by doing the manipulations yourself so you know exactly what is going on. The following R code adds those variables to the data.

```
balance$balance <- cumsum(c(0,
  balance$change[-nrow(balance)]))
balance$time <- 1:nrow(balance)
balance$flow <-
  factor(sign(balance$change))
```

To make the plot, we will first recreate the rectangles that show the change in the balance. In ggplot2, the geometric object used to represent the data is called a *geom*, and the geom used to draw rectangles is called *geom_rect*. The following code creates a new plot using the balance data, and then adds on a layer of rectangles, with one rectangle for each observation centered horizontally on time, positioned vertically to run between the current and previous balance. The fill color of the rectangle is mapped to the flow. This produces Figure 3. You can see that this displays the essence of the data, but is not formatted quite as we desire.

```
ggplot(balance) +
  geom_rect(aes(xmin = time - 0.45,
    xmax = time + 0.45,
    ymin = balance, ymax = balance +
    change, fill = flow))
```

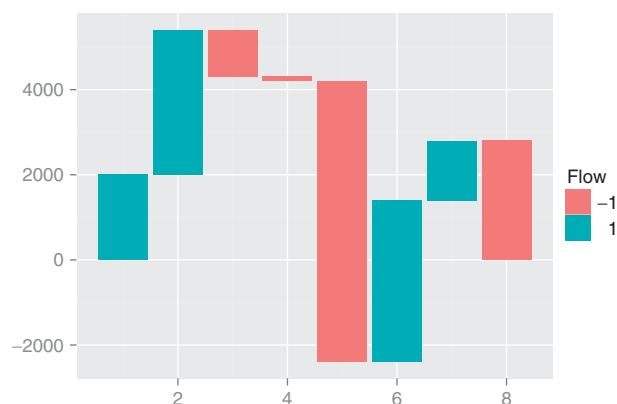


FIGURE 3 | Rectangles display change in balance at each time point.

Before we tweak the formatting, we need to add a layer of text, which is centered about time and placed at the bottom of the balance bar—the minimum of the previous balance and the new balance. We use the convenient dollar formatting function to make nicer labels. The results are shown in Figure 4.

```
ggplot(balance) +
  geom_rect(aes(xmin = time - 0.45,
    xmax = time + 0.45, ymin =
    balance, ymax = balance +
    change, fill = flow)) +
  geom_text(aes(x = time, y =
    pmin(balance, balance + change)
    - 50, label = dollar(change)),
    hjust = 0.5, vjust = 1,
    size = 3)
```

Now we need to focus on the presentation of the plot. During the exploratory phase many of the graphics you create will end up in the trash and you want tools that make iteration as fast as possible. You are already intimately familiar with the data,

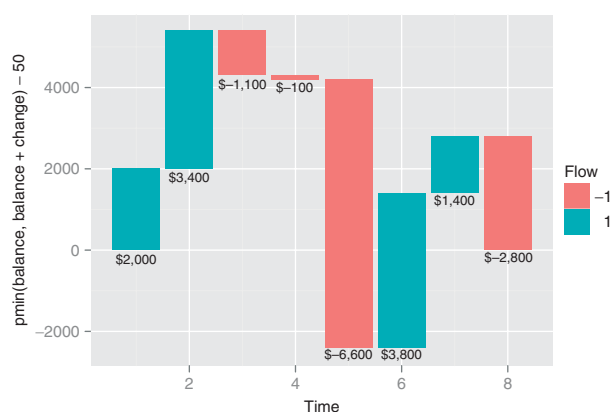


FIGURE 4 | Adding labels to show actual amounts.

so you do not need to fuss with nice labels, you just need the bare minimum of legends and axes to accurately decode the plot. As you transition toward the presentation/communication phase of your analysis, you want to spend more time making the plot easy to understand for those who are not already familiar with the data.

To improve the waterfall chart, we add an additional layer to show zero more clearly, but most of the work is on tweaking the scales. Scales are in charge of the mapping from data to perceptual properties, like position, shape, and color. Scales are also in charge of the inverse operation: creating the guides (legends and axes) which allow us to decode the data. Typical reasons for modifying the scale are to tweak the choice of outputs for the target medium (e.g. black and white printer, large scale display with projector), to meet existing conventions or standards, or to better match the data type with the scale type. Here we modify the scales to meet existing conventions as follows:

- x: put a tick at every time point, and label with the description;
- y: give a nicer title and format the values nicely;
- fill: make negative flows red and positive flows black, and do not display a legend (we use these colors to match existing financial standards to avoid problems for people with red-green color blindness).

```
ggplot(balance) +
  geom_hline(yintercept = 0, colour =
    "white", size = 2) +
  geom_rect(aes(xmin = time - 0.45,
    xmax = time + 0.45,
    ymin = balance, ymax = balance +
    change, fill = flow)) +
  geom_text(aes(x = time, y =
    pmin(balance, balance + change)
    - 50, label = dollar(change)),
    hjust = 0.5, vjust = 1,
    size = 3) +
  scale_x_continuous("", breaks =
    balance$time, labels =
    balance$event) +
  scale_y_continuous("Balance",
    formatter = dollar) +
  scale_fill_manual(values = c("-1"
    = "red", "1" = "black"),
    legend = F)
```

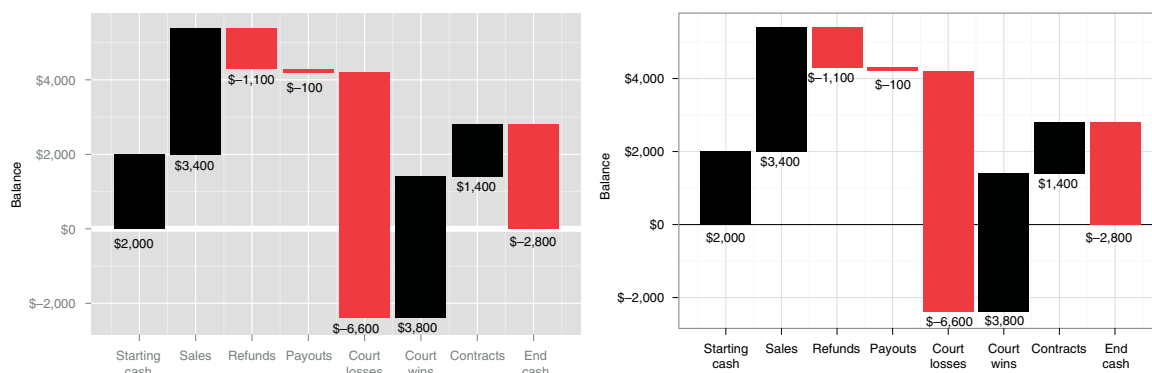


FIGURE 5 | Two examples of polished versions of the waterfall chart. (Left) Modifying the scales makes the plot easier to understand, and (right) modifying the theme gives a traditional white background with black gridlines.

Figure 5 shows the results of these operations. The second graphic shows an alternative representation that uses themes to modify the overall display. Themes control all nondata aspects of the plot, and are largely useful to override the default display properties of ggplot2 for meet publication requirements. The code to produce this graphic is included in the online supplementary materials.

PROGRAMMING GRAPHICS

The underlying grammar, another special feature of ggplot2 is that it is code based and programmable. This is typical for R, but unusual in general: most visualization tools use a graphical user interface. Programmability is important because it facilitates good science, which needs reproducibility, automation, and communication:

- If others cannot *reproduce* what you have done, then they are unlikely to believe your results. With R code, it is possible to reproduce an analysis exactly as it was performed previously.
- If reproducibility is about bringing the past into the present, *automation* is about bringing the present into the future. Most data sources are not static, but change over time, whether that is known in advance or as errors are fixed as analysis proceeds. A script that reproduces an analysis can be easily re-run whenever the data changes.
- Code also makes *communication* much easier because it can be copied and pasted; compare that with the difficulties of communicating a work flow in Excel.

One place that this is particularly useful is in graphical inference, a toolkit for checking that what

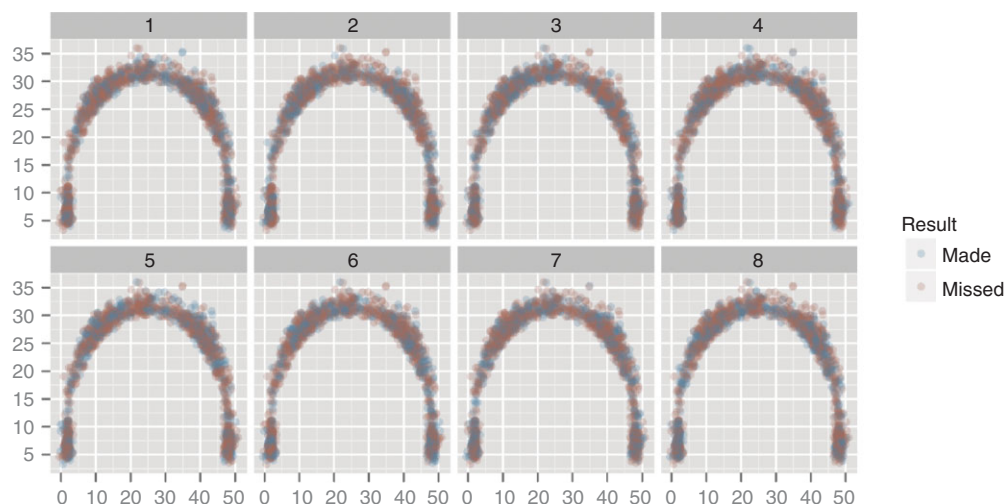


FIGURE 6 | Eight scatterplots show the location of three pointers attempted by the Los Angeles Lakers; the shots are colored by outcome. One plot shows the real data and seven plots show data from the null hypothesis. Can you spot the real data?

you see in a plot is really there.^{6,7} Figure 6 shows eight scatterplots that display the position of all three point shots attempted by the Los Angeles Lakers in the 2008/2009 season. Each attempt is colored by whether it was successful or not. This type of plot is called a lineup, because of the eight plots, only one is of the real data. The other seven were generated under the null hypothesis that position and success are unrelated. Can you pick out a plot that looks different? If you can, and it is the real data, then we have some evidence ($p = 1/8$) that there is a real difference. (Answer: The real data is in panel 2—did you spot it?)

The power of a programmatic approach to visualization is illustrated by the code used to create this plot, facilitated by the `nullabor` package^b. The essence of the code is shown below. Note the second line: it defines the method used to generate samples from the null hypothesis (permuting the result variable), and the total number of panels we want (8) in a natural extension to the grammar. This makes graphical inference easy to plug in to any existing visualization created with `ggplot2`.

```
ggplot(threep, aes(x, y)) %+%
  lineup(permute("result", n = 8) +
    geom_point(aes(colour = result,
      group = 1), alpha = 1/4) +
    scale_colour_manual(values =
      c("made" = mns1("10B 6/6"),
        "missed" = mns1("10R 6/6")))) +
  xlab(NULL) + ylab(NULL) +
  coord_equal() +
  facet_wrap(~ .sample, nrow = 2)
```

GGPLOT2 ECOSYSTEM

A vital part of `ggplot2` is the community and ecosystem that has built up around it, as well as the `ggplot2` website, which provides an online version of the

documentation, there are three main components to the `ggplot2` ecosystem: the mailing list, the wiki, and code contributions.

Mailing List

The mailing list is a key component of the `ggplot2` community, providing a friendly environment to ask questions about `ggplot2` and graphical presentation. As of November 2010, the mailing list had nearly 1200 members, and around 7000 messages. It is difficult to know exactly who the members are (most people sign up with gmail accounts), but a 2010 survey of the `ggplot2` mailing list revealed that users come from think tanks, non-profits (including the world health organization), independent consultancies, government agencies, federal labs, and companies, both big and small (including ebay, google, and facebook), and are using `ggplot2` to visualize data from ecology, ecophysiology, evolution, environmental science, toxicology, microbiology, pharmacology, medicine, epidemiology, global health, psychology, psychophysics, linguistics, energy, industrial engineering, technology, chemistry, physics, political science, business, finance, quality control, and market research.

Initially I invested a lot of effort to ensure that all questions on the list were answered. Recently, the volume of messages has become too high to sustain this effort, and the mailing list has become largely self-sustaining with most questions answered by other community members.

In addition, the mailing list, `ggplot2` also has a budding community on the programming question and answer site `stackoverflow`^c. At the time of writing, over 250 questions had been asked and answered on the site. One compelling feature of the `stackoverflow` is the automatically generated frequently asked questions page^d which algorithmically determines the most common questions.

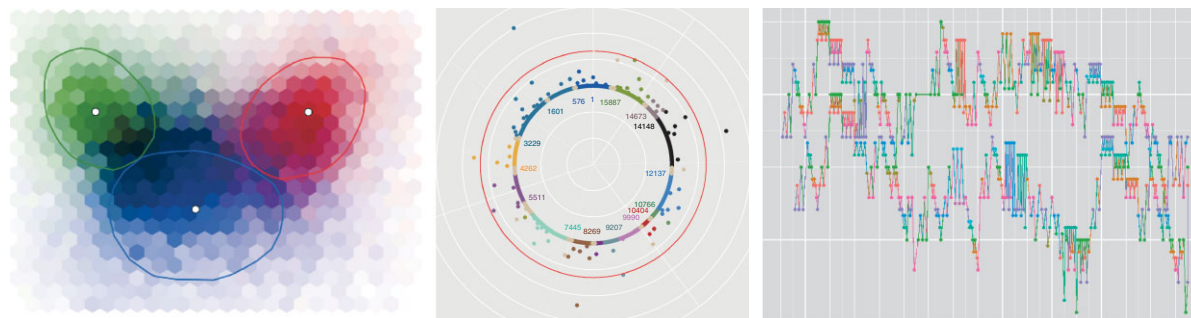


FIGURE 7 | Graphics created for the `ggplot2` case study competition for data on (left) the spectroscopic grading of gliomas, (center) mitochondrial DNA, and (right) the music of Bach.

Wiki

The chief feature of the ggplot2 wiki^e is the annual case study competition. This competition provides a venue for ggplot2 users to show off their graphics, and highlights how the richness of the grammar makes it possible to create a very wide range of graphics. Figure 7 shows parts of three entries from the 2010 competition.

The wiki also includes a section where users can list publications where they have used ggplot2, appearances of ggplot2 around the web, links to frequently asked questions and some tips and tricks.

Code Contributions

As the ggplot2 community grows, some users have begun to mature into developers and contribute code back to ggplot2. This is made possible, in part, by ggplot2's development process: everything occurs in the open using the code sharing site github, <http://github.com>. Github provides a number of services that make development easier including:

- Browsing the complete source, and history.
- Forking, which makes it easy for others to branch the main ggplot2 code and try out new things, and pull requests, which make it easy for me to merge these changes back into ggplot2 development.
- Line-by-line comments allow others to point out bugs that I have just introduced, and allow me to comment on other people's contributions.

CONCLUSIONS AND FUTURE WORK

ggplot2 implements a rich grammar of graphics in R that makes it possible to create new types of plot, each specifically tailored for the problem at hand. ggplot2 is designed to support reproducible research, making it possible to program graphics

making visualization reproducible, automated, and easy to communicate. This ease of communication has helped support the development of a strong community and rich ecosystem.

Currently, I am working on turning ggplot2 from the final product of my thesis research to fertile ground for future work. This involves breaking ggplot2 out into multiple simpler packages that each implement a core piece of the framework, like scales and layers, allowing the ggplot2 package to focus on the core grammar. Simpler packages will also make it easier for community members to transition from users to developers, allowing them to contribute new features or bug fixes in clearly defined areas. This work will also be useful for authors of other graphics packages, reducing the amount of duplicated code and making it easier to explore new ways of describing graphics, without having to devote so much time to fiddly lower-level details.

This work will also provide infrastructure for the next generation of graphics in R, which will combine the best of interactive and dynamic graphics from GGobi with the power and flexibility of a grammar. This is joint work with Michael Lawrence, Heike Hofmann, Di Cook, and the Iowa State statistical graphics research group. Initial exploratory work using an OpenGL based graphics device written by Michael Lawrence and Deepayan Sarkar has shown that it is possible to write performance interactive graphics in R.

NOTES

^a<http://learnr.wordpress.com/2010/05/10/ggplot2-waterfall-charts>, inspired in turn by <http://www.freakalytics.com/2009/11/17/wc/>

^bAvailable from <https://github.com/ggobi/nullabor>

^c<http://stackoverflow.com/tags/ggplot2>

^d<http://stackoverflow.com/tags/ggplot2/faq>

^e<https://github.com/hadley/ggplot2/wiki>

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